



Luca Lazzaroni

Assistant Professor in Electronics
DITEN – UNIGE

BASED IN Genoa, Italy
EMAIL luca.lazzaroni@unige.it

SECTOR RTD-A · IINF-01/A

BIBLIOMETRICS

SCOPUS

67 DOCUMENTS	338 CITATIONS	11 H-INDEX	20 ARTICLES, 5 YRS
------------------------	-------------------------	----------------------	------------------------------

SYNCED OCT 5, 2026

01 Profile

Assistant Professor (RTD-A) at the Department of Naval, Electrical, Electronic and Telecommunications Engineering (DITEN), University of Genoa, sector IINF-01/A – Electronics, since March 2025. PhD in Science and Technology for Electronic and Telecommunication Engineering since 2024.

Research on embedded machine learning and edge AI: TinyML toolchains, quantized and binary neural networks on microcontrollers and FPGAs, perception and decision-making for automated driving, deep reinforcement learning, on-device LLMs. The work is also carried out within European and national research projects with industrial partners.

Edge AI	TinyML	Embedded deep learning	Automated driving	Deep reinforcement learning
On-device LLMs				

02 Contact & profiles

EMAIL  luca.lazzaroni@unige.it	UNIGE  rubrica.unige.it	ORCID  0000-0001-8092-5473
SCOPUS  57220892898	GOOGLE SCHOLAR  w_6a6sYAAAAJ	GITHUB  @lucalazzaroni
LINKEDIN  luca-lazzaroni		

03 Appointments & education

POSITIONS

Mar 2025 – present	Assistant Professor (RTD-A) DITEN, University of Genoa Scientific sector IINF-01/A – Electronics.
--------------------	--

Oct 2023 – Feb 2025	Research Fellow DITEN, University of Genoa Grant: “Machine Learning models for the processing of data from autonomous vehicles”.
Nov 2020 – Oct 2023	PhD Candidate DITEN, University of Genoa — XXXVI cycle Scholarship: “Deep learning on the edge”.

EDUCATION

2024	PhD in Science and Technology for Electronic and Telecommunication Engineering University of Genoa — awarded March 14, 2024 Thesis: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning”
2020	MSc in Electronic Engineering (LM-29) University of Genoa — July 24, 2020 Thesis: “Design and implementation of a multiplatform execution engine for IoT applications” · Grade: 110/110
2017	BSc in Electronic Engineering and Information Technologies (L-8) University of Genoa — October 27, 2017 Thesis: “Model-based design of Lego robot control in Simulink environment”
2014	Scientific High School Diploma Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV) Grade: 100/100

Licensed Information Engineer — State Examination passed in the second session of 2020.

04 Teaching & academic appointments

COURSES

A.Y. 2024/25 – 2026/27	Distributed Electronic Systems and Technologies for Environmental Monitoring Course lecturer · cod. 94631
A.Y. 2024/25 – 2026/27	Tiny Machine Learning Co-lecturer · cod. 114752
A.Y. 2026/27	Electronic Devices and Circuits Course lecturer · cod. 65997
since A.Y. 2022/23	Machine Learning for Automated Driving Subject expert · cod. 108716
A.Y. 2023/24	Computer Graphics Arch. — Games and Simulation Adjunct professor · cod. 172317

A.Y. 2022/23	Autonomous Agents in Games Teaching support · cod. 198216
A.Y. 2020/21	Fundamentals of Programming of Electronic Systems Teaching support · cod. 166061
A.Y. 2020/21	Makers Approach to Electronic Design Teaching support · cod. 186985

All courses at the University of Genoa.

SUPERVISION

- 1** PhD candidate — Ali Zare, Science and Technology for Electronic and Telecommunication Engineering (XLI cycle)
- 12** BSc theses — Electronic Engineering and Information Technologies (L-8)
- 10** MSc theses — Electronic Engineering (LM-29)
- 7** MSc theses — Technological Engineering for Strategy and Security (LM/DS)
- 2** MSc theses — Computer Science (LM-18)

ACADEMIC APPOINTMENTS

Member of the Teaching and Admissions Committee, MSc in Engineering for Natural Risk Management.

Communications officer for the MSc in Electronic Engineering.

05 Editorial & service

EDITORIAL ROLES

2026	Volume Editor APPLEPIES 2026 — Springer LNEE Proceedings in preparation.
2025	Editorial Board Member (Early Career) Sensors — MDPI Since December 18, 2025.
2025	Volume Editor APPLEPIES 2025 — Springer LNEE Proceedings of the International Conference on Applications in Electronics Pervading Industry, Environment and Society.
2024	Guest Editor Energies — MDPI Special Issue “Tiny Machine Learning for Energy Applications”, April 2024 — August 2025.

CONFERENCE ORGANIZATION

Program Chair APPLEPIES 2026 Bologna, Italy, September 10–11, 2026.
Program Chair APPLEPIES 2025 Turin, Italy, September 11–12, 2025.
Program Committee APPLEPIES Since 2023.
Program Committee GALA — Games and Learning Alliance Since 2023.
Session Chair APPLEPIES 2023 1st Short Oral session — Genoa, Italy, September 28–29, 2023.

PEER REVIEW

JOURNALS	CONFERENCES
IEEE Access since 2023	IEEE ICRA — International Conference on Robotics and Automation since 2026
Journal of Systems Architecture since 2023	IEEE/RSJ IROS — Intelligent Robots and Systems since 2025
Engineering Applications of Artificial Intelligence since 2024	GALA — Games and Learning Alliance since 2023
Sensors since 2024	APPLEPIES since 2021
IET Intelligent Transport Systems since 2025	
IEEE Transactions on Intelligent Vehicles since 2026	
IEEE Transactions on Circuits and Systems II: Express Briefs since 2026	
Robotics and Autonomous Systems since 2026	
Scientific Reports since 2026	
Journal of Big Data since 2026	
Discover Internet of Things since 2026	

06 Invited & conference talks

13 TALKS

Sep 11–12, 2025	APPLEPIES 2025 Turin, Italy
Jan 11–14, 2025	43rd IEEE ICCE — International Conference on Consumer Electronics Las Vegas, NV, USA
Sep 19–20, 2024	APPLEPIES 2024 Turin, Italy
Jun 24–26, 2024	55th Annual Meeting of the Italian Society of Electronics (SIE) Genoa, Italy
Sep 28–29, 2023	APPLEPIES 2023 Genoa, Italy
Sep 7–9, 2023	54th Annual Meeting of the Italian Society of Electronics (SIE) Noto (RG), Italy
Sep 26–27, 2022	APPLEPIES 2022 Genoa, Italy
Sep 7–9, 2022	53rd Annual Meeting of the Italian Society of Electronics (SIE) Pizzo (VV), Italy
Sep 21–22, 2021	APPLEPIES 2021 Pisa, Italy
Dec 1–2, 2021	GALA 2021 — Games and Learning Alliance La Spezia, Italy (online)
Jul 7–9, 2021	52nd Annual Meeting of the Italian Society of Electronics (SIE) Trieste, Italy (online)
Dec 9–10, 2020	GALA 2020 — Games and Learning Alliance Laval, France (online)
Nov 19–20, 2020	APPLEPIES 2020 Online

07 Awards & memberships

2025	Best Presentation of the Session — “Edge Computing” 43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, January 11–14, 2025 A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti — “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”
------	---

PROFESSIONAL MEMBERSHIPS

IEEE

IEEE Member and IEEE Young Professional.

IEEE Consumer Technology Society

Member of the “Machine Learning, Deep Learning and AI in CE (MDA)” Technical Committee.

SIE – Italian Society of Electronics

Member within the University of Genoa Research Unit since 2021, research area “Electronic systems and applications”.

08 Research projects

Synergies

Generation of scenarios for development, training, virtual testing and validation of CCAM systems

European — HORIZON-CL5-2023-D6-01-02 · Task Leader, WP8 — Evaluation Framework (T8.1)

Generates test scenarios for training, testing and validating CCAM systems, focusing on urban and rural traffic — where relevant-scenario knowledge is far thinner than for highway driving. Runs with most major European carmakers and universities active in the field.

Hi-Drive

Addressing challenges toward the deployment of higher automation

European — H2020, GA 101006664 · Member of the University of Genoa research unit
Develops and tests advanced automated driving at scale, moving from conditional automation (SAE L3) to high automation across different scenarios and road contexts.

L3Pilot

Piloting Automated Driving on European Roads

European — H2020 · Member of the University of Genoa research unit
Large-scale piloting of SAE L3 automated driving functions on public roads across Europe, with the major European carmakers and suppliers.

GEMINI

Green Machine Learning for the IoT

National — PRIN 2022 · Member of the University of Genoa research unit (DII–UNITN / DITEN–UNIGE)

A framework for fully sustainable TinyML applications deployed along an edge-to-cloud continuum, targeting battery-free embedded devices — including those with no network connectivity at all.

ADINA

Integrated Digital Assistant Automotive

Regional — POR FESR Piedmont 2014-2020 · Member of the University of Genoa research unit, with Lifetouch

An in-vehicle virtual assistant on dedicated hardware, connected to the car's systems: AI algorithms interpret the driver's requests and act on vehicle and infotainment functions.

2026

Cross-platform deployment and characterization of an on-device generative conversational agentP. Firpo, R. Berta, F. Bellotti, **L. Lazzaroni**, M. Fresta, A. Pighetti, A. Abedinia, N. Yoonesi, D.P. Pau*Journal of Systems Architecture* vol. 181, art. 103981 Journal**A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction**L. Forneris, R. Berta, M. Fresta, **L. Lazzaroni**, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti*IEEE Signal Processing Letters* vol. 33, pp. 136–140 Journal**Efficient Certified Robustness Assessment for Transformer Models via Verifier-Aware Perturbation-Position Scheduling**V. Soltanmuradov, R. Berta, **L. Lazzaroni**, D.M. Gómez, A. Pighetti, F. Bellotti*Machine Learning and Knowledge Extraction* vol. 8, no. 9, art. 290 Journal**FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction**A. Saad, R. Berta, M. Fresta, **L. Lazzaroni**, A. Pighetti, O. Srouf, F. Tango, F. Bellotti*IEEE Transactions on Circuits and Systems II: Express Briefs* vol. 73, no. 8, pp. 888–892 Journal**Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms**H. Rojhan, F. Bellotti, M. Fresta, **L. Lazzaroni**, R. Berta*IEEE Access* vol. 14, pp. 44886–44903 Journal**StabilEdge, Formal Verification of Edge AI Applications****L. Lazzaroni**, R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti*IEEE Access* vol. 14, pp. 37416–37434 Journal**Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers**A. Saad, F. Bellotti, H. Ballout, M. Fresta, **L. Lazzaroni**, O. Srouf, R. Berta*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 321–327 Conference**Preface**M. Ruó Roch, R. Berta, A. Gómez, **L. Lazzaroni**, M. Martina, P.M. Ros, N. Pazos Escudero*Lecture Notes in Electrical Engineering* pp. v–vi Editorial**Optimizing Binary AutoEncoders on STM32 Microcontrollers**H. Ballout, R. Berta, **L. Lazzaroni**, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 301–307 Conference**Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students**L. Motta, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, N. Busi*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 242–248 Conference**Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform**H. Rojhan, F. Bellotti, **L. Lazzaroni**, M. Fresta, L. Forneris, A. Pighetti, R. Berta*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering pp. 218–223 Conference

Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA

D. Faryar, R. Berta, M. Fresta, A. Saad, **L. Lazzaroni**, H. Ballout, O. Srour, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 25–31

Conference

Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning

O.L. Audi, F. Bellotti, R. Berta, Z. Liu, **L. Lazzaroni**, A. Pilla, H. Rojhan, L. Forneris

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 212–217

Conference

Implementing YoloV8 on Zynq UltraScale + MPSoC FPGA Boards

H. Ballout, R. Berta, O. Srour, **L. Lazzaroni**, M. Fresta, A. Saad, F. Bellotti

Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 289–294

Conference

FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving

A. Saad, H. Ballout, F. Bellotti, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 295–301

Conference

Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision

A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 85–90

Conference

Embedded Deployment of an LLM-Based Voice Assistant

L. Lazzaroni, F. Bellotti, A. Pighetti, A. Saad, O. Srour, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 273–278

Conference

Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection

V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 286–291

Conference

A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors

S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 183–190

Conference

A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems

E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 249–254

Conference

A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents

A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 79–84

Conference

2025

An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection

L. Lazzaroni, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta

IEEE Sensors Journal vol. 25, no. 24, pp. 44859–44874

Journal

TinyML Acceleration with MAX78000

A. Dabbous, **L. Lazzaroni**, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta

Proceedings of SIE 2024 Lecture Notes in Electrical Engineering pp. 468–474

Conference

Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection

L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 156–163

Conference

Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking

H. Ballout, R. Berta, I. Bochenko, **L. Lazzaroni**, A. Dhonthi, V. Hashemi, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 387–394

Conference

Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking

F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonthi, V. Hashemi, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 139–147

Conference

RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset

R. Berta, F. Bellotti, M. Cossu, L. Forneris, **L. Lazzaroni**, A. Pighetti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 395–402

Conference

RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

IEEE Sensors Letters vol. 9, no. 9, art. 6010204

Journal

Quantization of MobileNetV2 for Resource-Constrained Microcontrollers

L. Lazzaroni, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 497–504

Conference

Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board

H. Ballout, R. Berta, A. Dabbous, M. Fresta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 242–249

Conference

On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection

M. Fresta, F. Bellotti, **L. Lazzaroni**, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudero, F. Tango, R. Berta

IEEE Open Journal of the Industrial Electronics Society vol. 6, pp. 1351–1369

Journal

Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices

A. Capello, R. Berta, M. Fresta, **L. Lazzaroni**, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 323–330

Conference

Green Machine Learning for the IoT (GEMINI): A Position Paper

L. Lazzaroni, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 339–346

Conference

Exploring Decision Transformer for Highway Automated Driving

L. Forneris, F. Bellotti, R. Berta, **L. Lazzaroni**, C. Oh

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 123–130

Conference

Deep Learning-Based Real-Time Driver Cognitive Distraction Detection

M. Fresta, F. Bellotti, I. Bochenko, **L. Lazzaroni**, G. Merlhiot, F. Tango, R. Berta

IEEE Access vol. 13, pp. 26589–26607

Journal

Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection

M. Cossu, R. Berta, M.A. Bagheri Orumi, **L. Lazzaroni**, J.L. Sauvaget, H. Touil, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 250–257

Conference

Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices

A. Dabbous, **L. Lazzaroni**, F. Sakr, R. Berta, M. Fresta, F. Bellotti

2025 IEEE International Conference on Consumer Electronics (ICCE)

Digest of Technical Papers IEEE International Conference on Consumer Electronics

Conference

Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification

A. Dabbous, R. Berta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 234–241

Conference

2024

An embedded end-to-end voice assistant

L. Lazzaroni, F. Bellotti, R. Berta

Engineering Applications of Artificial Intelligence vol. 136, art. 108998

Journal

Adversarial Policy Generation in Automated Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving

Workshop

CBin-NN: An Inference Engine for Binarized Neural Networks

F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Bellotti

Electronics vol. 13, no. 9, art. 1624

Journal

End-to-End Dataset Collection System for Sport Activities

M. Fresta, F. Bellotti, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Ansovini, R. Berta

Electronics vol. 13, no. 7, art. 1286

Journal

YoloP-Based Pre-processing for Driving Scenario Detection

M. Cossu, R. Berta, L. Forneris, M. Fresta, **L. Lazzaroni**, J.L. Sauvaget, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 418–423

Conference

Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection

M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 444–449

Conference

Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations

A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 124–129

Conference

Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity

R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti

Journal of Intelligent and Connected Vehicles vol. 7, no. 3, pp. 229–244

Journal

Developing a TinyML Image Classifier in an Hour

R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti

IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 946–960

Journal

Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving

L. Lazzaroni, A. Pighetti, F. Bellotti, R. Berta

Proceedings of SIE 2023 Lecture Notes in Electrical Engineering pp. 351–356

Conference

Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge

A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti

IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 781–794

Journal

Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach

L. Lazzaroni, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 352–357

Conference

Application of a Serious Game for Emotion Elicitation Under Socio-Economic and Trust Based Decision-Making Scenarios for Autistic Adolescents

F. Ahmed, R. Berta, F. Bellotti, F. Floris, **L. Lazzaroni**, G. Barresi, J.R. Carrion

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 170–180

Conference

2023

Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles

L. Forneris, A. Pighetti, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, R. Berta

International Journal of Serious Games vol. 10, no. 4, pp. 153–170

Journal

Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild

F. Ahmed, R. Berta, F. Bellotti, **L. Lazzaroni**, F. Floris, G. Barresi, J.R. Carrion

Applied Sciences vol. 13, no. 11, art. 6432

Journal

Supporting a .csv-based Workflow in MongoDB for Data Analysts

M. Fresta, A. Capello, F. Bellotti, **L. Lazzaroni**, M. Cossu, R. Berta

2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE)

IEEE International Symposium on Industrial Electronics

Conference

Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 307–311

Conference

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

IEEE Access vol. 11, pp. 28522–28550

Journal

Efficient Uploading of.Csv Datasets into a Non-Relational Database Management System

M. Fresta, F. Bellotti, A. Capello, M. Cossu, **L. Lazzaroni**, A. De Gloria, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 9–15

Conference

Developing a Toolchain for Synthetic Driving Scenario Datasets

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 222–228

Conference

Designing an Interpretability Analysis Framework for Deep Reinforcement Learning (DRL) Agents in Highway Automated Driving Simulation

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

Proceedings of SIE 2022 Lecture Notes in Electrical Engineering pp. 239–244

Conference

Deep Reinforcement Learning for Automated Car Parking

L. Lazzaroni, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 125–130

Conference

2022

Synthetic Driving Scenarios Generator

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

National

Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

National

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, R. Berta, A. De Gloria, A. Capello, M. Cossu, **L. Lazzaroni**

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

National

An Embedded End-to-End Voice Assistant

L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

National

Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications

R. Berta, F. Bellotti, A. De Gloria, **L. Lazzaroni**

Sensors vol. 22, no. 3, art. 713

Journal

High-Level Decision-Making Non-player Vehicles

A. Pighetti, L. Forneris, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 223–233

Conference

Developing a Synthetic Dataset for Driving Scenarios

J. Motta, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, S. Bonora

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 310–316

Conference

Classifying Simulated Driving Scenarios from Automated Cars

M. Cossu, J.L.Q. Villon, F. Bellotti, A. Capello, A. De Gloria, **L. Lazzaroni**, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 229–235

Conference

A Serious Game for Using Socio-Economic and Trust Based Decision-Making Scenarios for Elicitation of Emotional Responses

F. Ahmed, J.R. Carrion, F. Bellotti, **L. Lazzaroni**, G. Barresi, R. Berta

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 43–52

Conference

2021

Project and Development of an Italian Offline Embedded Voice Assistant

L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta

52nd Annual Meeting of the Italian Society of Electronics (SIE), Trieste

National

Edgine, A Runtime System for IoT Edge Applications

R. Berta, A. Mazzara, F. Bellotti, A. De Gloria, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 261–266

Conference

Adapting Autonomous Agents for Automotive Driving Games

G. Campodonico, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, T. Taccioli, F. Davio

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 101–110

Conference

2020

Employing an IoT Framework as a Generic Serious Games Analytics Engine

L. Lazzaroni, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 79–88

Conference